

Luke Guerdan

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Education

- 2021–Current **Carnegie Mellon University.**
Ph.D. Student in Human-Computer Interaction
Advisors: [Steven Wu](#), [Kenneth Holstein](#)
Courses: Machine Learning (PhD), Causal Inference, Reinforcement Learning, Behavioral Decision-Making, Modern Techniques in Uncertainty Quantification
- 2020–2021 **Cambridge University.**
MPhil in Advanced Computer Science (Honors with Distinction)
Advisor: [Hatice Gunes](#)
Thesis: "Federated Continual Learning for Human-Robot Interaction"
- 2015–2019 **University of Missouri.**
B.S. in Computer Science, Psychology (GPA: 4.0/4.0)
Advisors: [Yi Shang](#), [Tim Trull](#), [Steven Hackley](#)

Awards & Honors

- 2023 Best Paper Award at ACM Conference on Fairness, Accountability, and Transparency (FAccT)
- 2023 PhD Fellow, Center for Advancing Safety of Machine Intelligence (CASMI) at Northwestern University
- 2020 National Science Foundation Graduate Research Fellowship
- 2019 University of Missouri Award for Academic Distinction
- 2018 DAAD RISE Fellowship
- 2018 Barry Goldwater Scholarship in Science and Engineering

Academic Publications

[Google Scholar Link](#) (An asterisk (*) below indicates joint first-author.)

Refereed Conference & Journal Publications (Stringently Peer Reviewed)

- 2024 **Luke Guerdan**, Amanda Coston, Kenneth Holstein, and Zhiwei Steven Wu. Predictive performance comparison of decision policies under confounding. In *Proceedings of the International Conference on Machine Learning (ICML '24)*, 2024.
- 2023 Anna Kawakami, **Luke Guerdan**, Yang Cheng, Kate Glazko, Matthew Lee, Scott Carter, Nikos Archiga, Haiyi Zhu, and Kenneth Holstein. Training towards critical use: Learning to situate AI predictions relative to human knowledge. In *ACM Conference on Collective Intelligence (CI '23)*, 2023.
- 2023 **Luke Guerdan** and Hatice Gunes. Federated continual learning for socially assistive robotics. In *Proceedings of the 32nd IEEE International Conference on Robot & Human Interactive Communication (RO-MAN '23)*, 2023.

- 2023 **Luke Guerdan**, Amanda Coston, Zhiwei Steven Wu, and Kenneth Holstein. Ground(less) truth: A causal framework for proxy labels in human-algorithm decision-making. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*, 2023.
- 2023 **Luke Guerdan**, Amanda Coston, Kenneth Holstein, and Zhiwei Steven Wu. Counterfactual prediction under outcome measurement error. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*, 2023. **Best Paper Award**.
- 2019 Kendall Park, Kourtney Meiss, **Luke Guerdan**, Ev Cheng, Josiah Burchard, John Gillis, Prasad Calyam, and Salman Ahmad. Real-time geotracking and cataloging of mass casualty incident markers in a search and rescue training simulation: Pilot study. In *American Journal of Disaster Medicine*, 2019.
- 2019 Lukas Gehrke*, **Luke Guerdan***, and Klaus Gramann. Extracting motion-related subspaces from eeg in mobile brain/body imaging studies using source power comodulation. In *Proceedings of the IEEE/EMBS Conference on Neural Engineering (NER '19)*, 2019.
- 2018 William Morrison, **Luke Guerdan**, Jayanth Kanugo, Timothy Trull, and Yi Shang. Tigeraware: An innovative mobile survey and sensor data collection and analytics system. In *Proceedings of the IEEE Conference on Data Science in Cyberspace (DSC '18)*, 2018.
- 2017 **Luke Guerdan**, Olivia Apperson, and Prasad Calyam. Augmented resource allocation framework for disaster response coordination in mobile cloud environments. In *Proceedings of the IEEE Conference on Mobile Cloud Computing, Services, and Engineering (MobileCloud '17)*, 2017.
- 2016 Peng Sun, Nicholas Wergeles, Chen Zhang, **Luke Guerdan**, Timothy Trull, and Yi Shang. Ada: automatic detection of alcohol usage for mobile ambulatory assessment. In *Proceedings of the IEEE Conference on Smart Computing (SMARTCOMP '16')*, 2016.

Refereed Workshop & Poster Publications (Lightly Peer Reviewed)

- 2023 **Luke Guerdan**, Amanda Coston, Steven Wu, and Kenneth Holstein. Policy comparison under unmeasured confounding. In *NeurIPS 2023 Workshop on Regulatable Machine Learning*, 2023.
- 2022 Anna Kawakami, **Luke Guerdan**, Yang Cheng, Anita Sun, Alison Hu, Kate Glazko, Nikos Arechiga, Matthew Lee, Scott Carter, Haiyi Zhu, and Kenneth Holstein. Towards a learner-centered explainable AI: Lessons from the learning sciences. In *Workshop on Human-Centered Explainable AI (HCXAI) at the ACM CHI Conference on Human Factors in Computing Systems (CHI '22')*, 2022.
- 2022 **Luke Guerdan**, Kenneth Holstein, and Zhiwei Steven Wu. Under-reliance or misalignment? how proxy outcomes limit measurement of appropriate reliance in AI-assisted decision-making. In *Workshop on Trust and Reliance in AI-Human Teams (TRAIT) at the ACM CHI Conference on Human Factors in Computing Systems (CHI '22')*, 2022.
- 2022 **Luke Guerdan**, Amanda Coston, Ken Holstein, and Steven Zhiwei Wu. Ground(less) truth: The problem with proxy outcomes in human-AI decision-making. In *NeurIPS 2022 Workshop on Human-Centered AI (HCAI)*, 2022.
- 2022 **Luke Guerdan**, Amanda Coston, Ken Holstein, and Steven Zhiwei Wu. Counterfactual decision support under treatment-conditional outcome measurement error. In *NeurIPS 2022 Workshop on Causality for Real-world Impact*, 2022.
- 2021 **Luke Guerdan**, Alex Raymond, and Hatice Gunes. Toward affective XAI: facial affect analysis for understanding explainable human-AI interactions. In *Workshop on Responsible Pattern Recognition and Machine Intelligence (RPRMI) at the International Conference on Computer Vision (ICCV '21')*, 2021.

- 2019 **Luke Guerdan***, Peng Sun*, Connor Rowland, Logan Harrison, Zhicheng Tang, Nickolas Wergeles, and Yi Shang. Deep learning vs. classical machine learning: A comparison of methods for fluid intelligence prediction. In *Adolescent Brain Cognitive Development Neurocognitive Prediction: First Challenge, ABCD-NP 2019, Held in Conjunction with MICCAI 2019*. Springer International Publishing, 2019.

Research & Industry Experience

- 2021-Current **Carnegie Mellon University**, *Graduate Student*.
Advisors: [Kenneth Holstein](#), [Steven Wu](#).
Conduct research developing tools for measuring the reliability and validity of algorithmic systems.
- 2024 **Microsoft Research NYC**, *Ph.D. Research Intern*, Fairness, Accountability, Transparency & Ethics (FATE) Team.
Mentors: [Alexandra Chouldechova](#), [Solon Barocas](#), [Hanna Wallach](#).
Developed a statistical framework for principled evaluation of LLMs on subjective NLP tasks.
- 2018-2021 **TigerAware, LLC**, *Co-founder*.
Developed and commercialized mobile-based data collection platform for addiction research. Platform in use by 8,000+ research participants at over a dozen US research institutions, including Brown, Vanderbilt Medical Center, and Massachusetts General Hospital [[web](#), [press](#)].
- 2019 **Washington University in St. Louis**, *Summer Research Intern*, NSF REU in Big Data Analytics.
Advisor: [Chien-Ju Ho](#)
Investigated primal-dual based techniques for efficient online allocation of scarce societal resources.
- 2018 **Technische Universität Berlin**, *Summer Research Intern*, Berlin, Germany.
DAAD RISE Fellowship. Advisors: [Lukas Gehrke](#), [Klaus Gramann](#)
Developed semi-supervised learning algorithms for EEG-based neuroimaging.
- 2017 **ESRI**, *Software Engineering Intern*, Redlands, CA.
Developed geospatial database query optimization algorithms in an enterprise C++/COM stack.
- 2015-2019 **University of Missouri**, *Research Assistant*.
Advisors: [Yi Shang](#), [Tim Trull](#)
Investigated machine learning techniques for understanding human physiology and behavior.
Developed routing algorithms for real-time disaster response coordination (NSF REU in Consumer Networking Technologies, advised by [Prasad Calyam](#)).

Talks & Speaking Engagements

- 2024 **Invited Speaker**. INFORMS 24' Session on Human-Centered AI and Decision Making for Social Good. Seattle, WA. '*A Methodological Framework for Human-Algorithm Performance Comparisons Under Incomplete Information*.' (Upcoming October 2024)
- Invited Speaker**. EAAMO Meetup at ICML 24'. '*Counterfactual Prediction Under Outcome Measurement Error*.'
- Invited Speaker**. BIRS Workshop on Bridging Prediction and Intervention Problems in Social Systems. Banff International Research Station, Calgary CA. '*A Framework for Human-Algorithm Performance Comparison Under Incomplete Information*.'
- Invited Speaker**. Complexity Science Hub Vienna, Lecture Series on Network Inequality. Remote. '*Human-Algorithm Decision-Making Under Imperfect Proxy Labels*.'
- 2023 **Podcast Guest**. Still Updating Podcast with Nathan Gould. Remote. '*Finding Truth in Labeled Data*.'

- 2022 **Invited Speaker** (joint with Anna Kawakami). INFORMS 22' Session on Human-AI Teams. Indianapolis, IN. *'Understanding Human-AI Decision-Making in the Real-World: From Observational Studies to Theoretical Models.'*
- 2018 **Invited Speaker.** RISE 2018 Intern Summit. Heidelberg, DE. *'Extracting Motion-Related Sub-spaces from EEG in MOBI Studies.'*

Workshops Organized

- 2024 Jiaee Cheong, Nikhil Churamani, **Luke Guerdan**, Tabitha Edith Lee, Zhao Han, Hatice Gunes, Causal-HRI: Causal Learning for Human-Robot Interaction, HRI.

Grants

- 2017-2018 **University of Missouri Interdisciplinary Innovations Fund.**
\$25,000. Co-PI with William Morrison.

Service

Reviewer.

- 2024 FAccT, DIS
- 2023 ICLR, AAAI, CHI, FAccT, ACM Journal on Responsible Computing
- 2022 CSCW, FAccT

Departmental Service.

- 2023- Co-organizer of Fairness, Explainability, Accountability, and Transparency (FEAT) reading group.
- 2021-2023 Graduate Student Assembly (GSA) Departmental Representative.
- 2021-2023 Carnegie Mellon Graduate Application Support Program Mentor

Leadership.

- 2015-2019 Mizzou Computing Association. *President, Vice President, Machine Learning SIG Director.*
- 2017 TigerHacks. *Assistant Director.*
- 2019-2020 Show Me Dharma Buddhist Society. *Board Member, Director of Communications.*

Mentorship

- 2023 Ming Wang, High School Student, Summer Research Internship.
- 2022 Mahika Varma, Undergraduate, HCI Independent Study.
- 2022 Angelica Bonilla, Undergraduate, CMU AI Mentorship Program.
- 2021 Matthew Ok, Undergraduate, HCI Independent Study.